

CLASSIFICATION

FEATURE SELECTION

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FEATURE SELECTION

The following concepts will be introduced:

- ✓ **REDUNDANT/IRRELEVANT ATTRIBUTE**
- ✓ **FEATURE SELECTION APPROACHES**
- ✓ **FILTER**
 - UNI-VARIATE
 - MULTI-VARIATE
- ✓ **WRAPPER**
- ✓ **FEATURE SELECTION PROCEDURE FLOWCHART**
- ✓ **FEATURE CREATION**

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107	26	161.6	n	3	3.7	OH	415	371-7191
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147	0	157	n	6	1.92	MO	415	329-9001
117	0	184.5	n	4	2.35	KS	408	335-4719
141	37	258.6	n	5	3.02	WV	415	330-8173



Your friend is happy with the **CLASSIFIER YOU DEVELOPED** to solve the **CHURN PROBLEM**. Indeed, it incurs an **EXPECTED COST WHICH IS SMALLER THAN THAT INCURRED BY THE STANDARD MAILOUT PROCEDURE**.

However, your friend asks you to check whether **THE 17 INPUT ATTRIBUTES** used by the Classifier **ARE ALL USEFUL TO SOLVE THE CHURN PROBLEM**. Indeed, your friend wishes to **REDUCE THE NUMBER OF INPUT ATTRIBUTES TO IMPROVE THE CLASSIFIER'S INTERPRETABILITY**.

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You make the decision to **DESIGN AND APPLY TO THE CUSTOMERS CALLS DATABASE** a **FEATURE SELECTION PROCEDURE** (Attribute Selection) **TO DISCOVER WHICH ATTRIBUTES ARE:**

- ✓ **REDUNDANT:** contains information already available due to one or more other attributes.
- ✓ **IRRELEVANT:** contains information not useful to solve the considered data mining task.

FEATURE SELECTION: **APPROACHES**

1

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Different **APPROACHES** are available:

- ✓ **BRUTE-FORCE**: to apply all possible subsets of the available attributes as the input attributes to the Classification Model.

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available attributes

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available attributes

5

possible subsets

31

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5
10

possible subsets

31
1,023

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$$\sum_{n=1}^{10} \binom{10}{n}$$

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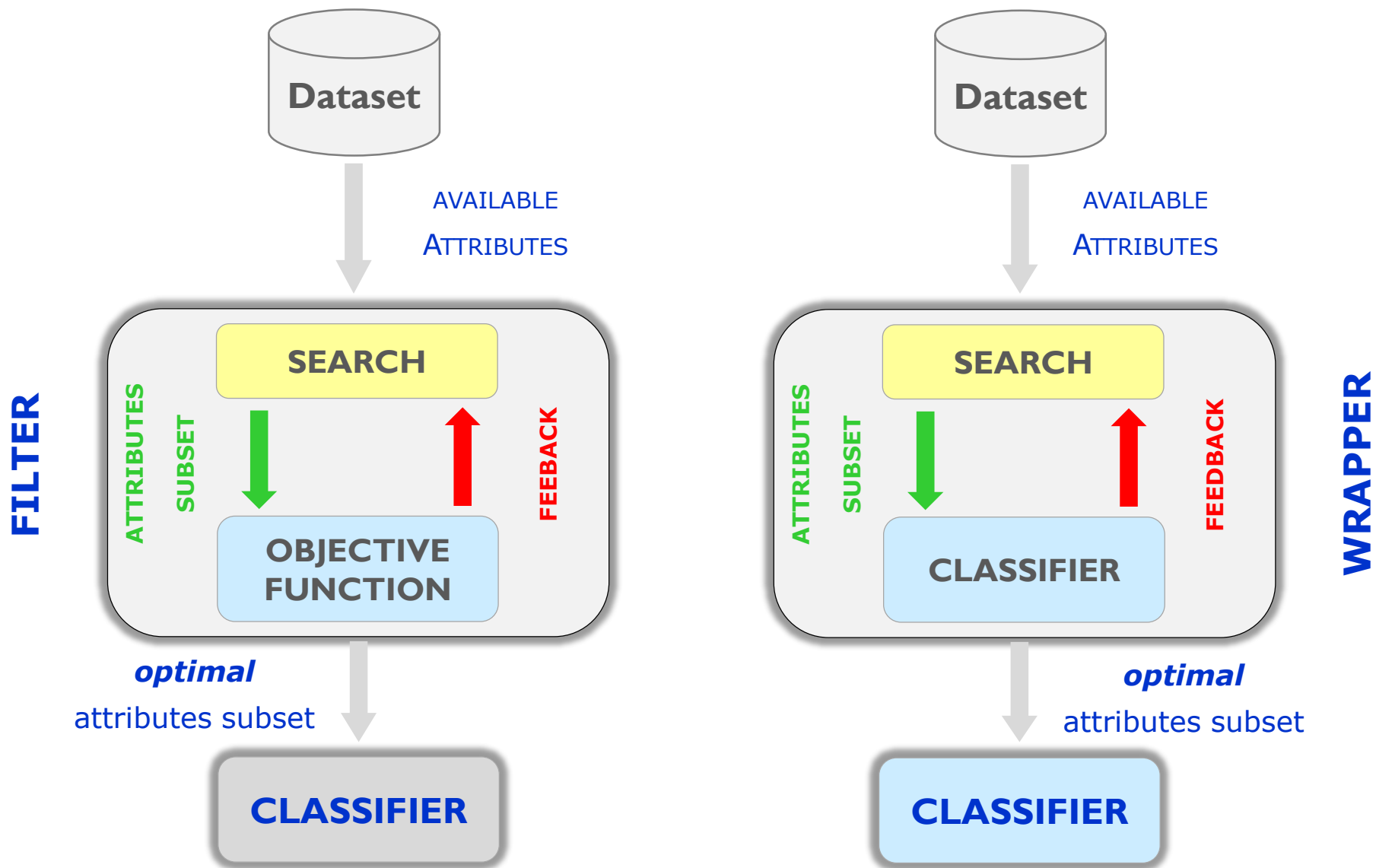
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- ✓ **EMBEDDED**: attributes selection occurs as a by-product of Classifier Learning (BN, DT, ...).
- ✓ **FILTER**: attributes are selected before learning the Classifier.
- ✓ **WRAPPER**: the Classifier is used to find the optimal subset of the available attributes.

FEATURE SELECTION: FILTER & WRAPPER

2



Filters can be:

✓ **UNI-VARIATE**

- Choose an **ASSOCIATION MEASURE BETWEEN CANDIDATE ATTRIBUTES AND CLASS ATTRIBUTE**, i.e., mutual information, gain ratio, ...
- **SORT** the candidate Attributes **ACCORDING TO THE ASSOCIATION MEASURE**.
- **SELECT THE «R» BEST CANDIDATE ATTRIBUTES** as Input Attributes for Classification learning.
- It is likely that **IRRELEVANT ATTRIBUTES** are correctly identified (removed) while **REDUNDANT ATTRIBUTES** are not correctly identified (not removed).

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✓ **MULTI-VARIATE**

- **JOINTLY IDENTIFY IRRELEVANT AND REDUNDANT ATTRIBUTES**
- A good subset of attributes contains **ATTRIBUTES WHICH ARE STRONGLY ASSOCIATED WITH THE CLASS ATTRIBUTE BUT UNCORRELATED AMONG THEM**
- Symmetric uncertainty measures or correlation-based measures

FEATURE SELECTION: **UNI-VARIATE & MULTI-VARIATE**

4

Uni-Variate

Multi-Variate

Uni-Variate	Multi-Variate
<i>Parametric</i>	
t-test	
ANOVA	
Mutual Information	

Uni-Variate	Multi-Variate
<i>Parametric</i>	
t-test	
ANOVA	
Mutual Information	
<i>Non Parametric</i>	
Mann-Whitney	
Kruskal-Wallis	
Permutation test	

Uni-Variate	Multi-Variate
<i>Parametric</i>	Correlation Feature Selection (CFS) Relief Blanket
t-test	
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	Advantages	Disadvantages
Uni-Variate	speed scalability independent on the classifier	ignore that attributes can be dependent ignore interactions with the classifier

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Mutual Information	
Non Parametric	
Mann-Whitney	
Kruskal-Wallis	
Permutation test	

	Advantages	Disadvantages
Uni-Variate	speed scalability independent on the classifier	ignore that attributes can be dependent ignore interactions with the classifier
Multi-Variate	model dependency between attributes independent on the classifier computational cost compares favorably to wrapper	slower than Uni-Variate techniques less scalable than Uni-Variate Techniques ignore interactions with the classifier

The main **ADVANTAGES** are:

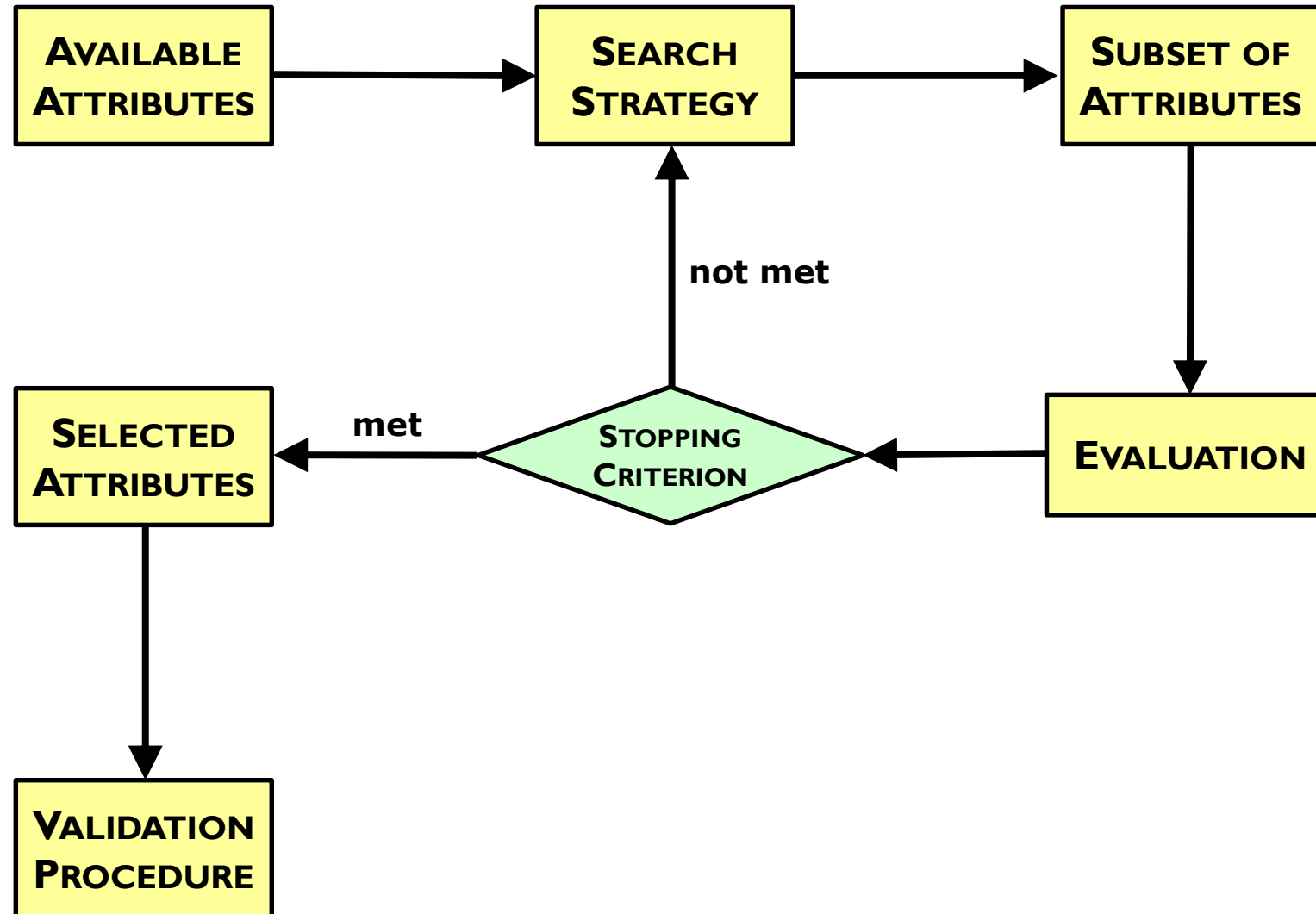
- ✓ Reduction of the cost of data collection
- ✓ Reduction of the inference time, i.e., time required by the Classifier to predict the value of the Class Attribute
- ✓ Increased interpretability
- ✓ Increased Accuracy

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The main **MOTIVATIONS** are:

- ✓ to **AVOID THE OVERFITTING PHENOMENA**
- ✓ to develop a faster and **COST-EFFECTIVE CLASSIFIER**
- ✓ to improve the **UNDERSTANDING OF THE DATA GENERATING PROCESS**



It is often possible to create, from the original attributes, a **NEW SET OF ATTRIBUTES** that **CAPTURES THE RELEVANT INFORMATION IN A DATA SET** much more effectively.

Furthermore, the **NUMBER OF NEW ATTRIBUTES CAN BE SMALLER THAN THE NUMBER OF ORIGINAL ATTRIBUTES**, allowing you to reap the benefits of **DIMENSIONALITY REDUCTION**.

Some **METHODOLOGIES FOR CREATING NEW ATTRIBUTES** (features) are:

- ✓ **FEATURE EXTRACTION**; a much broader set of classification models can be applied, they are highly domain-specific (image processing)
- ✓ **MAPPING THE DATA TO A NEW SPACE**, a totally different view of the data can reveal interesting features which increase understanding and classification performance
- ✓ **FEATURE CONSTRUCTION**; the available attributes contain information that is not suitable for some classification algorithms, one or more new features constructed out of the original attributes can be more useful than the original features